
📘 AWS CloudFormation – Overview

🎯 What is AWS CloudFormation?

AWS CloudFormation is an **Infrastructure as Code (IaC)** service that lets you **automate the creation, update, and deletion** of AWS resources using code (called **templates**).

> 🔧 Instead of manually creating AWS resources, you define them in a **YAML or JSON template**, and CloudFormation provisions everything for you.

🏗️ Key Concepts

Term	Description
Template	A YAML or JSON file describing the desired AWS infrastructure
Stack	A group of AWS resources created from a CloudFormation template
Resources	Actual AWS services like EC2, S3, IAM, etc. defined in the template
Parameters	Inputs to customize template behavior
Outputs	Values returned after stack creation (e.g., VPC ID, URL)
Mappings	Static lookup tables inside templates
Conditions	Logic to include/exclude resources based on parameters
Metadata	Additional data about resources or templates

🛠️ How It Works

1. **You write a template** in YAML or JSON

2. **Upload it to CloudFormation**
3. **CloudFormation reads the template** and creates the resources as a **stack**
4. If you update the template, CloudFormation **updates the stack automatically**
5. Delete the stack → **all resources are deleted**

✅ Benefits

Feature	Benefit
Infrastructure as Code	Automate and version-control infrastructure
Repeatability	Create identical environments (e.g., Dev, Test, Prod)
Dependency Handling	Automatically provisions resources in the correct order
Rollback Support	Automatically reverts if something fails
Drift Detection	Detects manual changes to stack resources
Integration	Works with CodePipeline, CI/CD, AWS SAM, CDK

📦 Supported Resources

CloudFormation supports **almost all AWS services**, including:

- EC2, VPC, S3, IAM, RDS, Lambda
- Route 53, CloudFront, ELB, Auto Scaling, and more

📄 Sample Template (YAML)

```yaml
AWSTemplateFormatVersion: '2010-09-09'
Description: Simple EC2 instance

Resources:
  MyInstance:
    Type: AWS::EC2::Instance
    Properties:
      InstanceType: t2.micro
      ImageId: ami-0abcdef1234567890
```

🧑‍💻 Use Cases

- 🔁 Repeatable environment deployment
- 🔄 Dev/Test/Prod provisioning
- 🧪 Testing infrastructure changes before production
- 🏗️ Infrastructure version control (with Git)

⚠️ CloudFormation vs Other IaC Tools

Tool	Language	Notes
CloudFormation	YAML/JSON	Native AWS support
Terraform	HCL	Multi-cloud support
AWS CDK	TypeScript/Python/Java	High-level abstraction over CloudFormation

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💬 Summary

Feature	Details
What	IaC service for managing AWS infrastructure
How	Define resources in YAML/JSON, launch stacks
Why	Automates, scales, and secures infrastructure deployment
Cost	Free (you pay only for the resources created)
